

Requirements Analysis (Ambiguity & Conflicts)

In this phase, we analyzed the collected requirements to identify and resolve any potential ambiguities or contradictions. This ensures the system is logical, consistent, and technically feasible.

Ambiguity Resolution

- Ambiguity 1: Real-Time Tracking Precision (Ref: FR-06)
 - Issue: The term "Real-time" is technically vague. Does it mean millisecond updates or minute-by-minute?
 - Resolution: We defined "Real-time" as GPS location updates sent every 5 seconds, with status changes reflected on the user interface within a maximum latency of 2 seconds.
- Ambiguity 2: Proximity for "Nearby" Restaurants (Ref: FR-02)
 - Issue: "Nearby" varies based on urban density and delivery logistics.
 - Resolution: We specified a default search radius of 10km, which can be dynamically adjusted by the system administrator based on driver availability in specific zones.

Conflict Resolution

- Conflict 1: High Security vs. Seamless Usability (Ref: FR-01 & Usability)
 - Issue: Strong security (multi-factor authentication/re-entering card details) conflicts with the goal of placing an order in under 5 taps.
 - Resolution: We implemented Secure Tokenization for payments. This allows users to pay with saved "tokens" (One-tap payment) while the actual sensitive data is handled securely by a third-party PCI-DSS compliant gateway.
- Conflict 2: Driver Proximity vs. Service Quality (Ref: FR-09)
 - Issue: Assigning the "Nearest Driver" might lead to a poor experience if that driver has a low rating.
 - Resolution: The assignment algorithm was refined to prioritize drivers based on a weighted score of both Proximity and Rating, ensuring the fastest delivery without compromising service quality.